Computer Engineering Department,S V N I T, Surat.
End-Semester Examination, May 2018

B Tech IV(CO)–8th semester
Course: (CO-428) Cloud Computing (EIS-II)

Date: 5th May ,2018 **Time: 15:30 to18:30** **Max Marks: 50**

Instructions:

1. Write your B Tech Admission No/Roll No and other details clearly on the answer books and write your BTech Admission No on the question paper, too.
2. Assume any necessary data but give proper justifications.
3. Be precise and clear in answering the questions.

Q1. Answer the following. (Any 5) **[30]**

1. Discuss Application Delivery Controller (ADC) in detail.
2. Several desirable properties of a large-scale distributed system include transparency of access, location, concurrency, replication, failure, migration, performance, and scaling. Analyze how each one of these properties applies to Amazon web services (AWS).
3. Virtualization simplifies the use of resources, isolates users from one another, supports replication and mobility, but exacts a price in terms of performance and cost. Analyze each one of these aspects for:
   (i) Memory virtualization
   (ii) Processor virtualization
   (iii) Network virtualization
4. Compare the three cloud computing delivery models, SaaS, PaaS, and IaaS,from the point of view of the application developers and users. Discuss the security and the reliability of each one of them. Analyze the differences between the PaaS and the IaaS.
5. Compare cloud services provided by Amazon, Google, and Microsoft.
6. Discuss VMware vSphere in detail.

Q2. Answer the following. **[20]**

1. What are the cloud computing security risks from Gartner's perspective?
2. Justify your answers. (Reference: Paper discussed in class: Exploring information leakage in third-party compute clouds)

   Q1: Can one determine where in the cloud infrastructure an instance is located?

   Q2: Can one easily determine if two instances are co-residence on the same physical machine?

   Q3: Can an adversary launch instances that will be co-residence with other user's instances?

   Q4: Can an adversary exploit cross-VM information leakage once co-resident?
3. Network security is one of the key point in cloud security. How network security works from cloud security perspective?
4. Describe computing scaling hardware service management in cloud computing.

*****